Project: Worked examples
Title: Beam formulas (single span)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Closed-form reactions, maximum moment and its position, maximum shear and maximum deflection for the standard single-span beam cases, as a quick check. Left end at x = 0; loads downward; sagging moments positive; deflection downward positive.

WHAT IT CALCULATES	RELATION
Formulas from Roark Table 8.1, in this sheet's own notation, for the chosen support and load case.	
Simply supported beam, point load at a (b = L − a), reaction and largest moment	$R_A = Pb/L$; $M_{max} = Pab/L$
Simply supported beam, point load at a, largest deflection when a ≥ b	$\delta_{max} = Pb(L^2 - b^2)^{(3/2)}/(9\sqrt{3}\,L\,EI)$ at $x = \sqrt{((L^2 - b^2)/3)}$
Propped cantilever, uniform load, reaction and moments	$R_B = 3wL/8$; $M_A = -wL^2/8$; sagging $M_{max} = 9wL^2/128$ at $x = 5L/8$

Basis: Shear, moment, slope and deflection formulas for elastic straight beams[1](Ch. 8, Table 8.1).

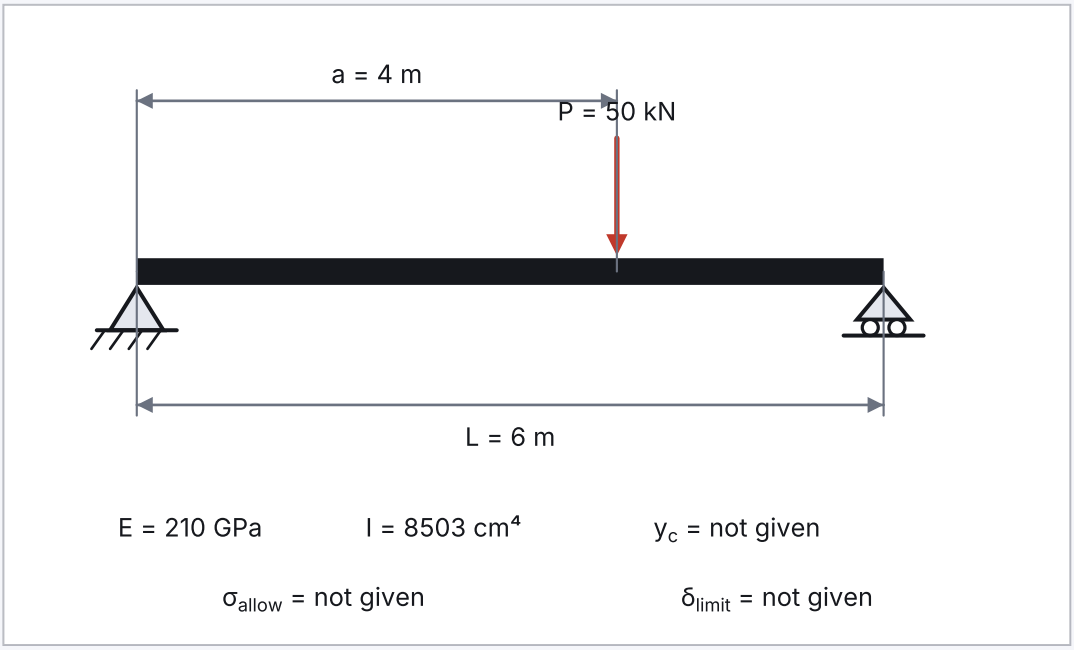


Figure 1 — Simply supported, point load P at a from the left end. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
case	Case	simply supported, point load at a		assumption: Single span, pin–roller.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L	Span	6	m	drawing: DRG-FL-201 rev A	0.1 … 100 m
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	1 … 500 GPa
I	Second moment of area	8503	cm^4	reference: UB 305×165×40 (Blue Book)	1 … 100000000 cm^4
P	Point load (downward)	50	kN	spec: Load schedule LS-01: plant load	0 … 100000 kN
a	Load position from the left end	4	m	drawing: DRG-FL-201 rev A	0 … 100 m

3. Assumptions

A1. Prismatic, linear elastic, Euler–Bernoulli beam; small deflections. (Roark Table 8.1 conditions.)
A2. A cantilever is fixed at the left end (x = 0) and free at the right. (Case convention.)
A3. case (Case) is taken as simply supported, point load at a. (Single span, pin–roller.)

4. Working

[5] $EI = E * I$
$= 210\ GPa \times 8503\ cm^4$
$= 17856\ kN \cdot m^2$
Flexural rigidity

[8] $b = L - a$
$= 6\ m - 4\ m$
$= 2\ m$
Distance from the load to the right end

[9] $R_A = P * b / L$ [1](8.1 2a)
$= 50\ kN \times 2\ m / 6\ m$
$= 16.667\ kN$
Left reaction

[10] $R_B = P * a / L$
$= 50\ kN \times 4\ m / 6\ m$
$= 33.333\ kN$
Right reaction

[11] $M_{max} = P * a * b / L$
$= 50\ kN \times 4\ m \times 2\ m / 6\ m$
$= 66.667\ kN \cdot m$
Maximum moment, under the load

[12] $x_M = a$
$= 4\ m$
$= 4\ m$
Position of the maximum moment

[13] $M_{min} = 0\ kN*m$
$= 0\ kN \cdot m$
Minimum moment

[14] $V_{max} = max(R_A, R_B)$
$= max(16.667\ kN, 33.333\ kN)$
$= 33.333\ kN$
Maximum shear

[15] c_{far} = b
= 2 m
= 2 m
Shorter segment (the deflection formula's b)
[16] x_{delta} = sqrt((L^2 - c_{far}^2) / 3)
= sqrt(((6 m)^2 - (2 m)^2) / 3)
= 3.266 m
Position of the maximum deflection
[17] δ_{max} = P * c_{far} * sqrt((L^2 - c_{far}^2)^3) / (9 * sqrt(3) * L * EI)
= 50 kN × 2 m × sqrt(((6 m)^2 - (2 m)^2)^3) / (9 × sqrt(3) × 6 m × 17856 kN·m²)
= 10.839 mm
Maximum deflection

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	16.667 kN	Left reaction
R_B	33.333 kN	Right reaction
M_{max}	66.667 kN·m	Maximum sagging moment
x_M	4 m	Position of M_{max}
M_{min}	0 kN·m	Minimum (hogging) moment
V_{max}	33.333 kN	Maximum shear
δ_{max}	10.839 mm	Maximum deflection
x_{delta}	3.266 m	Position of δ_{max}

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

No warnings.

8. Verification

Template struct.beam-formulas@1.0.1 Content hash 9ce95307c4fbf44b Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.